M. ARAVIVAT

S.No	Dates	Title.	Sign.
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2.	24/07/25	NETWORK CABLES	31/7/25 (10)
3.	31/08/25	PACKET TRACER TOOL INSTALLATION AND USER INTERFACE	21/8/25 (10)
4.	4/08/25	WIRESHARK.	12/8/25 (10)
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7.	8/9/25	Setup LAN using Switch and Ethernet	
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9.	29/9/25	Subnetting in cisco packet Tracer	
10.	3/10/25	Internetworking with Routers in cisco packet Tracer Simulator.	8/10/25 (10)
11.	5/10/25	Simulate static Routing configure using cisco packet Tracer.	
12.	7/10/25	a) Simulate static Routing protocol Config using cisco packet Tracer b) Simulate RIP using cisco	
13.	8/10/25	Implement your own ping prog ~~& chat prog using socket~~.	9/10/25 (10)
14.		RAW Sockets to implement- packet sniffing.	
15.		Analyse various types of servers using Metasploit tool	

Exp: 1

21/07/25 CS19541 - COMPUTER NETWORKS

AIM
Exp1 :- Study of Various Network Commands used in LINUX and Windows.

BASIC NETWORKING COMMANDS (windows)

① arp -a : shows IP address of your computer along with the IP and MAC address of your route

OUTPUT :-

Interface : 172.16.75.34 --- 0x7

Internet Address	Physical Address	Type
172.16.72.1	7c-5a-1c-cf-be-4d	dynamic
172.16.72.42	cc-47-40-8d-0d-af	dynamic

② hostname : Displays the name of your computer

output :

DESKTOP - 8938VQ1

③ ipconfig /all : Displays important configuration information about your TCP/IP connection.

output :

Windows IP Configuration

Host Name . . . . . : DESKTOP - 8938VQ1
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid.
:

⑨ Router : Views or modifies the IP routing table used to add / Delete static routes.

OUTPUT :-

Interface List

16 .... 20 88 10 86 7a be .... Intel (R) Ethernet Connecti

2 .... 4e 82 a9 78 96 e7 .... Microsoft Wifi Direct Virtual Adapter

SOME IMPORTANT LINUX NETWORKING COMMANDS.

① Ip : Basic Command Every administration will need in daily work.

→ ip address show.

OUTPUT :-

1. lo : <LOOPBACK, UP, LOWER_UP>

2. enp0s31f6 : .....

3. wlp2s0 ....

inet 172.16.75.125/21

→ Sudo ip address add 192.168.1.254/24 dev enp0s3

OUTPUT:

Connection established.

You are now connected to enp0s31f6

→ sudo ip address dell 192.168.1.254/24 dev enp0s31f6

→ sudo ip link set0 enp0s3if6 up

→ sudo ip link set enp0s3if6 down

→ sudo ip link set enp0s3if6 promise

→ sudo ip route add default via 192.168.1.254 dev enp0s13f6

② ifconfig.

The ifconfig cmd is a staple in many sysadmin's tool belt for configuring and troubleshooting Networks

OUTPUT

enp0s3if6 : flags = 4355 <UP, BROADCAST ...>

inet 192.168.1.100

lo : flags = 73 <UP, LOOPBACK, RUNNING>

inet 127.0.0.1

wlp2s0 : flags = 4163 <....>

inet 172.16.75.125

③ mtr

NTR (Matt's traceroute) is a program with a command line interface that serves as a Network diagnostic and troubleshooting tool. This command combines the functionality of the ping and traceroute commands

Syntax of the command

mtr < options > <hostname / IP>

common use cases

a) mtr google.com.

OUTPUT

Host	packets		pings			
	Loss %	snt	Last	Avg	Best	Wors
1. -gateway.	0.0%	85	2.4	7.5	1.7	79.1
2. static_41.229.249 ...	0.0%	85	66.9	14.0	4.4	132
⋮	⋮		⋮			

b) mtr -n google.com

Host	packets		pings		
	Loss %	snt	Last	Avg	Best...
1. 172.16.72.1	11.9%	163	8.3	28.2	2.4
2. 49.249.229.41	11.6%	66	6.7	41.6	4.9
⋮	⋮		⋮		

c) mtr -b gogögle.com.

d) mtr -c 10 google.com.

OUTPUT

Host	packets		pings			
	Loss %	snt	Last	Avg	Best	Worst
1. -gateway-	0.0%	2	6.2	8.9	6.1	239.4
2. static_41.229.249.	0.0%	2	5.5	11.0	22.1	72.2
⋮			⋮			

1) tcdump

The tcdump cmd is designed for capturing and displaying packets

→ Install tcpdump with the command :

1. sudo dnf Pnstall -y tcdump

OUTPUT :

:

package tcpdump-14:4.99.4-2-fc39.x86 is already installed.

pependencies resolved.

Nothing to do

Complete !

2. tcpdump -D

OUTPUT :

:

1. wlp2s0 [UP, Running, Wireless, Associated]
2. Any
3. lo [UP, Running, Loopback]
4. enpos31f6 [up, Disconnected]

:

3. sudo tcpdump -i wlp2s0

:

listening on wlp2s0...

:

5 packets captured,

481 packets received by

358 packets dropped by kernel.

:

4. sudo tcpdump -i wlp2s0 -c 10

OUTPUT:

Listening on wlps2s0, link-type EN10MB

:

10 packets captured.

39 packets received by filter

0 packets dropped by kernel.

5) To find traffic coming from and going to 8.8.8

use the command:

tcpdump -i wlp2s0 -c 10 host 8.8.8.8.

OUTPUT:

01:43:22.402561 IP fedora > dns.google ...

:

10 packets captured

10 packets received by filter.

0 packets dropped by kernel

6) For traffic coming from 8.8.8.8,

sudo tcpdump -i wlp2s0 src host 8.8.8.8.

OUTPUT:

Listening on wlp2s0, link type EN10MB...

23:21:22.453708 IP dns.google > fedora...

23.21.23.479039 IP dns.google...

:

5) ping :-

It is used to troubleshoot connectivity, reachability & name resolution. ping google.com

Output : ping google.com (142.253.221.288) 56 (34) bytes of data.

from fedora (193.168.8.294) icmp-seq = 1 Destination Host unreachable

from fedora (192.168.1.294) icmp-seq = 2. Destination Host unreachable.

Student Observation :

1) Which command is used to find the reachability of a host machine from your device

Ans: ping → used to check the reachability of a host

ping <hostname or IP>

ping google.com.

2) Which command will give the details of hops & IP taken by packet to reach the destination

Ans: traceroute → shows the path & hops a packet takes to reach a destination.

3) Which command displays the IP configuration of your machine

Ans: ifconfig or ip address → Displays IP configuration

4) Which cmd displays the TCP port status.

Ans:- netstat -t or ss -t → shows TCP port status

5) Write the command to modify the IP Configuration in a Linux Machine?

Ans: To add : ip address add 192.168.1.254/24 dev enp0s3

To del : ip address del 192.168.1.254/24 dev enp0s3

Result:-

Therefore, the study of various Network Command used in Linux and Windows is Executed Successfully

21/8/24

24/7/25
Exp-2:

NETWORK CABLES

Aim :- Study of Different types of Network Cables.

a). understand Different types of Network Cable.

Different types of cable used in networking are:

1. Unshielded Twisted pair (UTP) Cable
2. Shielded Twisted pair (STP) Cable.
3. Coaxial Cable
4. Fibre Optic Cable.

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application / use.
UTP	category 3	10 Mbps	Adv. • Cheaper in cost • Easy to install. Disadv. • Susceptible to EMI and noise.	• 10Base-T Ethernet • Fast Ethernet. • Gigabit Ethernet • ~~Fast E~~
	category 5	up to 100 Mbps		
	category 5e	1 Gbps		
STP	category 6, 6a.	10 Gbps	Adv • Shielded • Faster than UTP Disadv • Expensive • Greater installation Effort.	• Gigabit Ethernet, 10 G Ethernet • widely used in data centres
SSTP	Category 7	10 Gbps		
Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Adv • High bandwidth Disadv • Limited Distance & cost.	Speed of signal is 500m Television network. High speed Internet connection
fibre optics cable	Single mode Multi mode	100 Gbps	Adv • High Speed, Security, • Long Distance Disadv • Req skilled installer	Maximum Distance of fibre optic cable is around 100 meters

~~b) Make Your~~

STUDENT OBSERVATION

1. what is the Difference btw cross cable and straight

Ans Straight Cable : Same wiring on both ends (used for different devices)

cross cable : Different wiring on each end (used for similar devices)

2. which type of cable is used to connect two PCs?

Ans Cross Cable

3. which type of cable is used to connect router/switch to PC?

Ans: Straight Cable

4. Find out the category of twisted pair cable used in your LAN to connect the PC to the network socket.

Ans: Category 5e (cat 5e), or Category 6 (cat 6)

5. Write down your understanding, challenges faced, and o received while making a twisted pair cross / straight

Ans
- understanding : Arranged 8 wires in T568A/B std & cr into RJ-45.
- challenges : wire alignment, correct order, & proper
- output : Cable worked successfully & passed LAN Test

Result:

Thus, the Study of different type of Network Cable done successfully.

3/8/26

Exp-3

PACKET TRACER TOOL INSTALLATION AND USER INTERFACE

Aim: To study the packet tracer Installation and user Interface Overview.

a) Install and set up cisco packet Tracer.

b) Analyse the behaviour of Network devices using CISCO PACKET TRACER Simulator.

1) From the Network component box, click and drag-and-drop the below components:
   a) A Generic PCs and One HUB
   b) A Generic PC's and One Switch

2) Click on Connections:
   a) Click on Copper Straight-Through Cable,
   b) Select one of the PC and connect it to HUB using the cable. The link LED should glow in green, indicating that the link is up. Similarly connect remaining 3PCs to the HUB
   c) Similarly connect 4PCs to the switch using copper straight-through cable.

3) Click on the PCs connected to hub, go to the Desktop tab, click on IP Configuration, and enter an IP address and Subnet mask. Here, the default gateway and DNS server information is not needed as there are only two end devices in the Network.

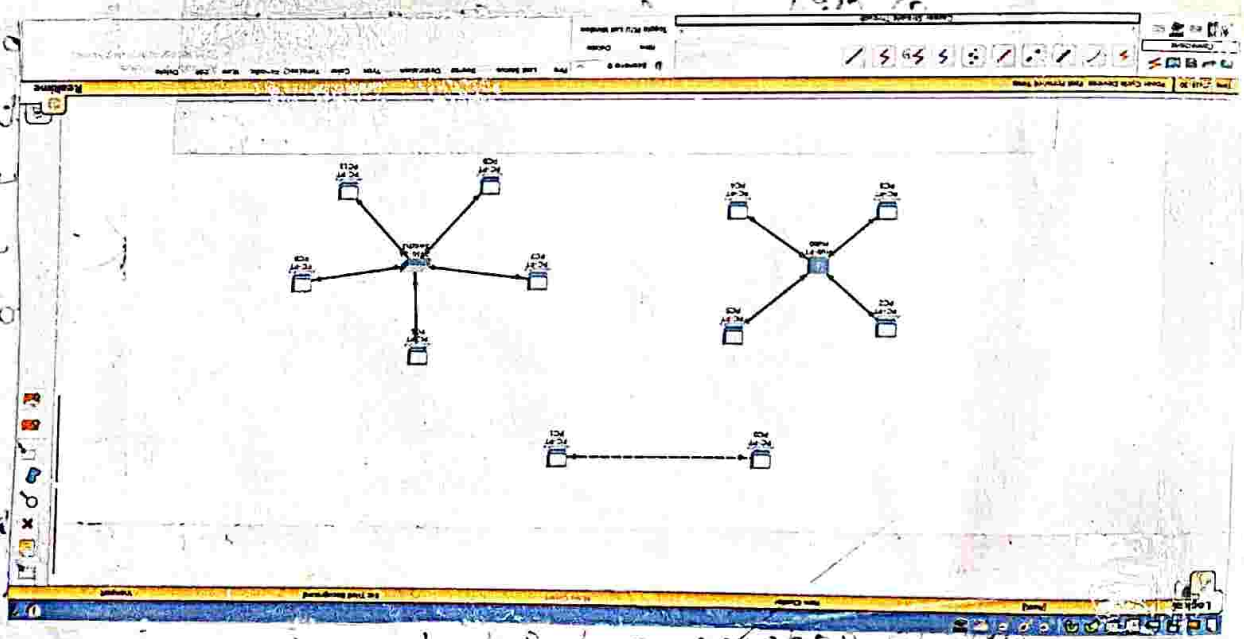
Desktop → IP config → Inter IP add. 10.10.10.1 & Subnet Mask. Click on the PDU from ~~sow~~ the common tool bar

a) Drag and drop it on one of PC (source machine) and then drop it on another PC (Destination machine) connected to the HUB.

4) Observe the flow of PDU from source PC to destination PC by selecting the Realtime mode of simulation.

5) Repeat step #3 to step #4 for the PCs connected to the switch

6) Observe how HUB and switch are forwarding the PDU and write your observation and conclusion about the behaviors of switch and HUB



STUDENT OBSERVATION:

a) From your observation write down the behavior of Switch and HUB in terms of forwarding the packets recieved by them.

Ans: Hub : Broadcasts packets to all ports, causing unnecessary traffic.

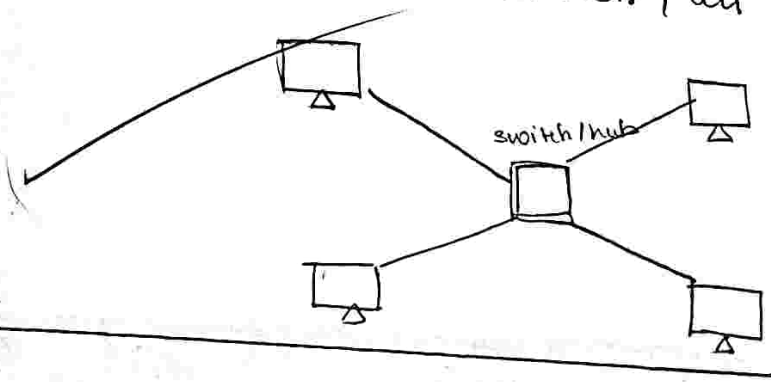
Switch: Fromwards packets only to the destination port using MAC address, reducing traffic.

b) Find out the Network topology implemented in your college and draw and label that topology in your observation book.

Ans: Network topology

Type : Star topology.

Drawing : One Central Switch, all Computers connected to it



Result :

The packet tracer tool Installation and User Interface Overview are Successfully Executed.

9/8/25

4/8/25

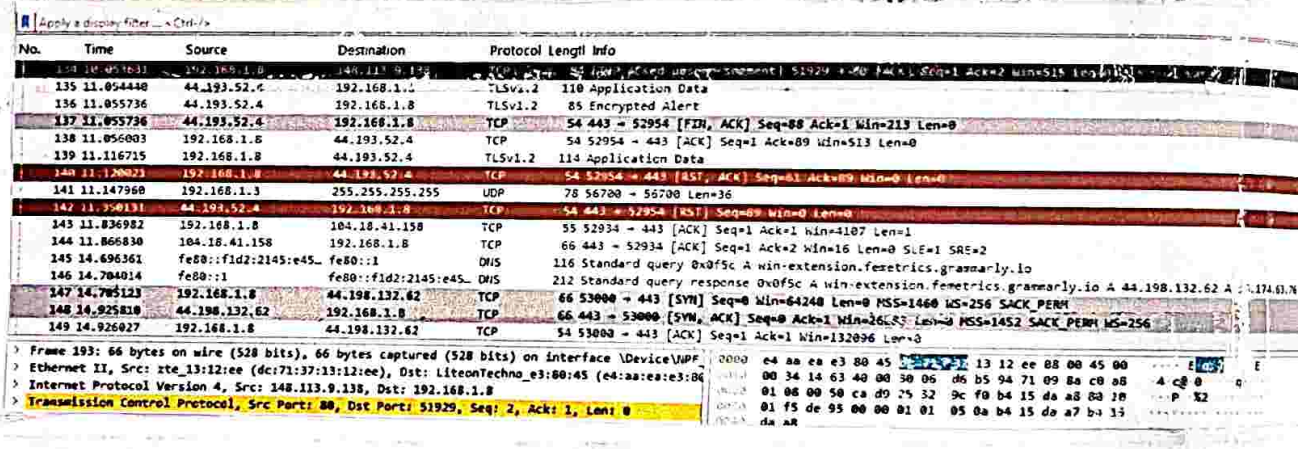
Exp: 4

WIRESHARK.

AIM: Experiments on Packet Capture tool: Wireshark

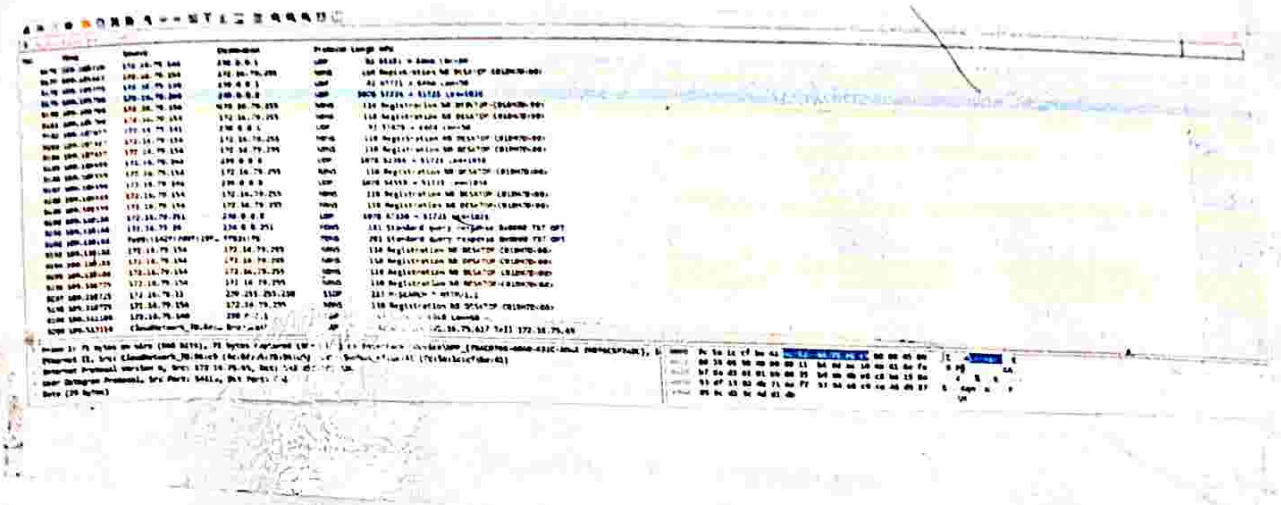
1. Create a Filter to display only TCP / UDP packets, Inspect the packets, and provide the flow graph

- Select Local Area Connection in wireshark
- Go to capture → option.
- Select stop capture automatically after 100 packets.
- Then click Start Capture.
- Search TCP packets in search bar.
- To see flow graph click Statistics → flow graph.
- Save the packets.



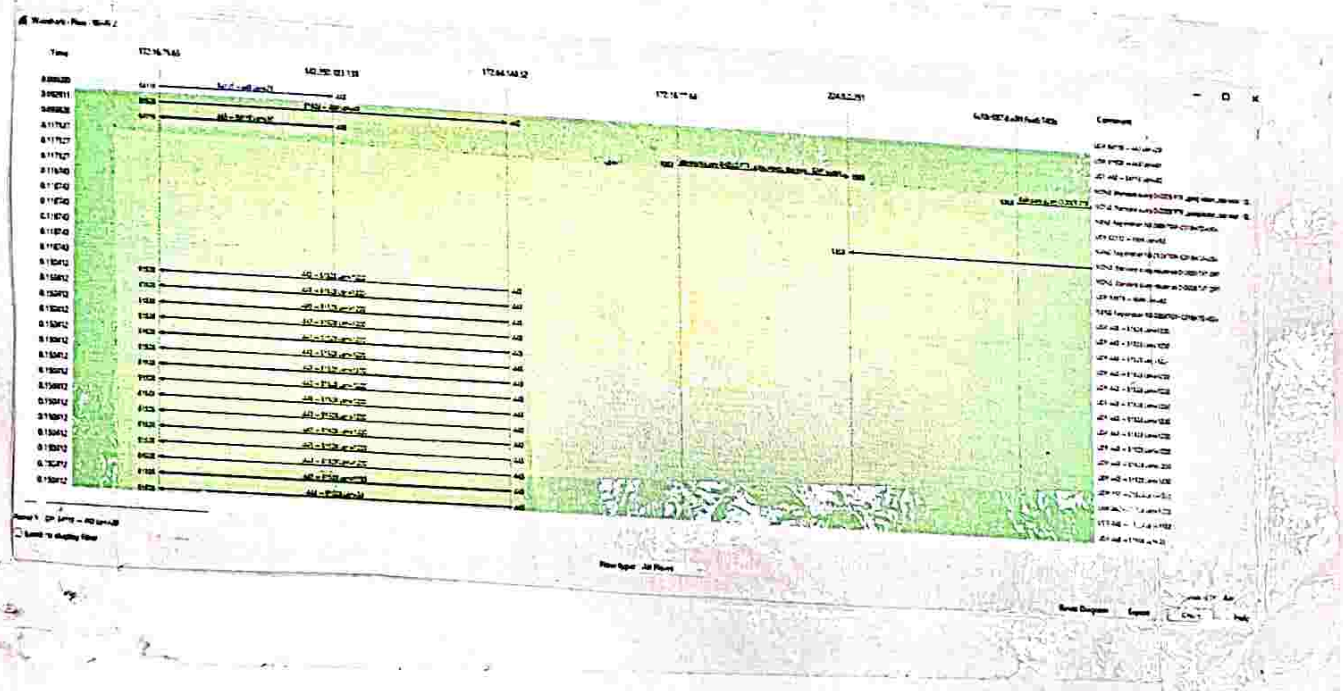
2. Create a Filter to display only ARP packets and Inspect the packets.

- Go to Capture → option
- Select stop capture automatically after 100 packets
- Then click start capture.
- Search ARP packets in search bar.
- Save the packets.



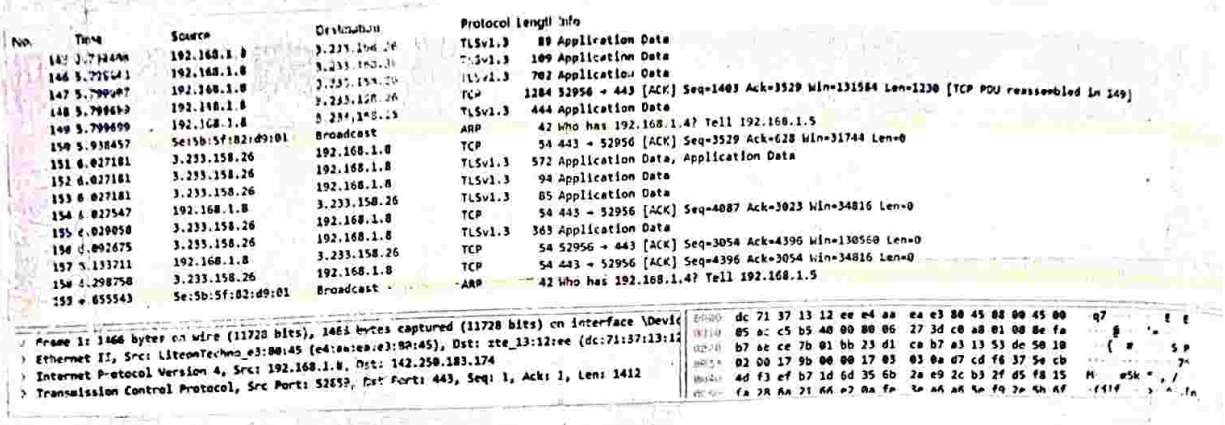
3. Create a Filter to Display only DNS packets and provide the flow graph.

- Go to capture → option
- ~~Select stop~~ Search DNS packets in search bar



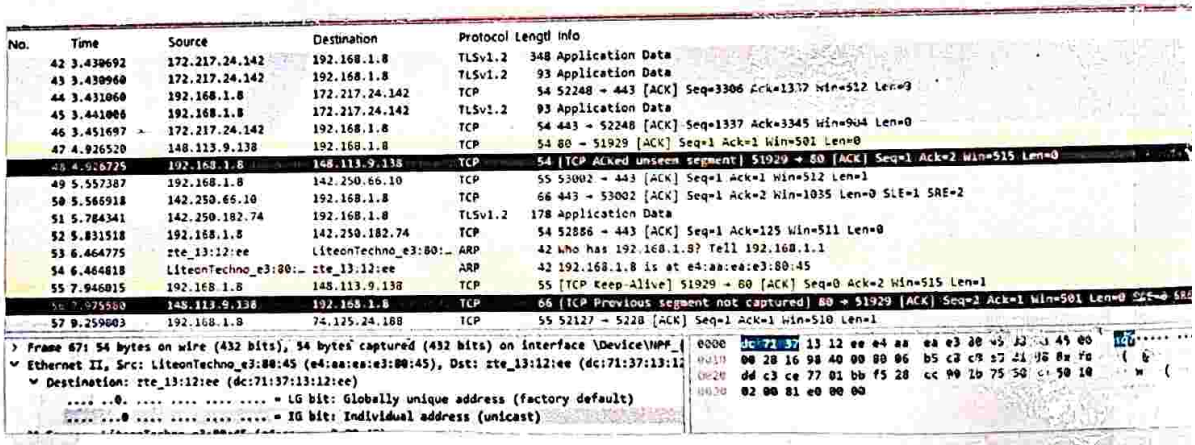
4. Create a Filter to display only HTTP packets and Inspect the packets

- Select Local Area Connection in Wireshark
- Search HTTP packets in Search bar.



5. Create a Filter to display only IP/ICMP packets and Inspect the packets

- Select Local Area Connection in Wireshark.
- Search ICMP/IP packets in search bar.



6. Create a Filter to display only DHCP packets Inspect the packets.

- Select Local Area Connection in Wireshark.
- Search DHCP packets in search bar.

Student Observation.

1) What is promiscuous mode ?
   - It is a network mode where a computer's network card captures all traffic (not just the traffic meant for it.
   - useful for tools like Wireshark to analyze all network packets.

2) Does ARP packets have a transport Layer header? Explain
   - No transport layer (TCP/UDP) header.
   - works at the data link layer.

3) ~~DNS~~ Which Transport layer protocol is used by DNS ?
   - UDP (port 53) for most queries (faster, no connection)
   - TCP (port 53) for larger data like zone transfers.

4) What is the port number used by HTTP protocol.
   The HTTP port number is 80.

5) What is a broadcast IP address ?
   - Sends to all devices in the Network.
   - Eg :- 192.168.1.255 in Local Network.

RESULT :-

The experiment was successfully performed using Wireshark.

18/8

1/9/25

Exp: 5

HAMMING CODE

AIM:

Write a program to implement error detection and correction using HAMMING code concept. Make a test run to input data stream and verify error correction feature.

1) Sender program.

- Convert input text to binary.
- Applies Hamming code to the binary data by adding redundant parity bits
- Saves the Encoded data to a file called channel

2) Receiver program.

- Reads the data from channel.
- checks for errors using hamming code.
- If an error is detected, It shows the position of the error
- If no errors, It removes redundant bits and converts the binary data back to ASCII

CODE

```
import java.util.Scanner;
public class HammingCode {
    static void print(int ar[]) {
        for (int i = 1; i < ar.length; i++) {
            System.out.print(ar[i]);
```

```
    }
    System.out.println();
}
static int[] calculation(int[] ar, int r) {
    for (int i = 0; i < r; i++) {
        int x = (int) Math.pow(2, i);
        for (int j = 1; j < ar.length; j++) {
            if (((j >> i) & 1) == 1) {
                if (x != j)
                    ar[x] = ar[x] ^ ar[j];
            }
        }
        System.out.println("r" + x + " = " + ar[x]);
    }
    return ar;
}

static int[] generateCode(String str, int m, int r) {
    int[] ar = new int[r + m + 1];
    int j = 0;
    for (int i = 1; i < ar.length; i++) {
        if ((Math.ceil(Math.log(i) / Math.log(2)) ==
            Math.floor(Math.log(i) / Math.log(2))) == 0) {
            ar[i] = 0;
        } else {
            ar[i] = (int) (str.charAt(j) - '0');
            j++;
        }
    }
    return ar;
}
```

```
public static void main (String [] args) {

    Scanner sc = new Scanner ( System.in );
    System.out.print (" Enter binary data : ");
    String str = sc. nextLine ();
    int M = str. length();
    int r = 1;

    while (Math.pow (2,r) < (M+r+1)) {

        r++;
    }
    int[] ar = generateCode (str, M, r);
    System.out.println (" In Generated Hamming Code:
    ar = calculation (ar, r);
    print (ar);
    }
}
```

OUTPUT

Enter Binary data (e.g. 1011) : 1111

Generated Hamming code :

r1 = 1

r2 = 1

r4 = 1

1 1 1 1

Result :-

Hamming Code Concept is Executed Successfully

1/9/25
Exp: 6

SLIDING WINDOW

AIM:

Write program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

~~SENDER~~ HOW TO USE:

- Inputs window size & message.
- Sends window_size frames at a time
- Writes frames to sender_Buffer.
- Receiver reads frames, ~~to~~ Sends ACK or NACK to Receiver_Buffer.
- Sender reads ACK / NACK and continues or resends ~~respon~~ frames
- You can Manually Edit the files to simulate errors.

CODE

```
import time.
import random.
class Sender:
    def __init__(self, total_frames, window_size):
        self.total_frames = total_frames
        self.window_size = window_size
        self.base = 0
        self.next_seq = 0
    def send_frames(self):
        print(f"\n[Sender] Total frames to send:
                {self.total_frames}")
```

```
        while self.base < self.total_frames:
            while self.next_seq < self.base + self.window_size and self.next_seq < self.total_frames:
                print(f"[Sender] Sending Frame {self.next_seq}")
                self.next_seq += 1
            time.sleep(1)

    def ack_received(self, ack):
        print(f"[Sender] Acknowledgment received for Frame {ack}")
        if ack >= self.base:
            self.base = ack + 1

class Receiver:
    def receive_frame(self, frame_no, sender):
        if random.choice([True, False]):
            print(f"[Receiver] Received Frame {frame_no}")
            sender.ack_received(frame_no)
        else:
            print(f"[Receiver] Frame {frame_no} lost! No Ack sent.")

if __name__ == "__main__":
    total_frames = 5
    window_size = 3
    sender = Sender(total_frames, window_size)
    receiver = Receiver()
    sender.send_frames(receiver)
```

OUTPUT

```
Enter total number of frames : 5
Enter window size : 3
[Sender] Total Frames to send : 5
[Sender] Sending frame 0
[Sender] Sending frame 1
[Sender] Sending frame 2.
[Receiver] Successfully received frames 0 to 2.
[Sender] Acknowledgment received for frame 2
[Sender] Sending Frame 3
[Sender] Sending Frame 4
[Receiver] Frame 4 lost or corrupted
[Sender] ⏱ Timeout ! Resending window from Frame 3
[Sender] Sending Frame 3
[Sender] Sending Frame 4.
[Receiver] Successfully received frames 3 to 4.
[Sender] Acknowledgment received for Frame 4
Transmission completed.
```

Result :- Sliding window protocol is Executed Successfully

3/9/26

15/9/25

Exp: 7

Setup & Configure a LAN using Switch & Ethernet Cables.

AIM: Setup and Configure a LAN using a Switch and Ethernet Cables in your Lab.

Step 1:- Plan & Design an appropriate Network topology taking into account Network requirem & Equipment location.

Step 2:- You can take and Computers, a Switch with 8, 16 or 24 ports which is Sufficient for Networks of these sizes & 4 Ethernet Cabl

Step 3: Assign IP Addresses

- On each PC :
  - > Go to Network & Internet Settings → Ethernet → Properties.
  - > Select IPV4 → Use the following IP address
  - > Set IP addresses as :
    - ↳ PC 1 : 10.1.1.1, Subnet Mask : 255.0.0.0
    - ↳ PC 2 : 10.1.1.2, Subnet Mask : 255.0.0.0
    - ↳ PC 3 : 10.1.1.3, Subnet Mask : 255.0.0.0
    - ↳ PC 4 : 10.1.1.4, Subnet Mask : 255.0.0.0

Step 4: Configure the Switch

- Connect one PC to the Switch.
- Open a browser → Enter Switch IP - log in.

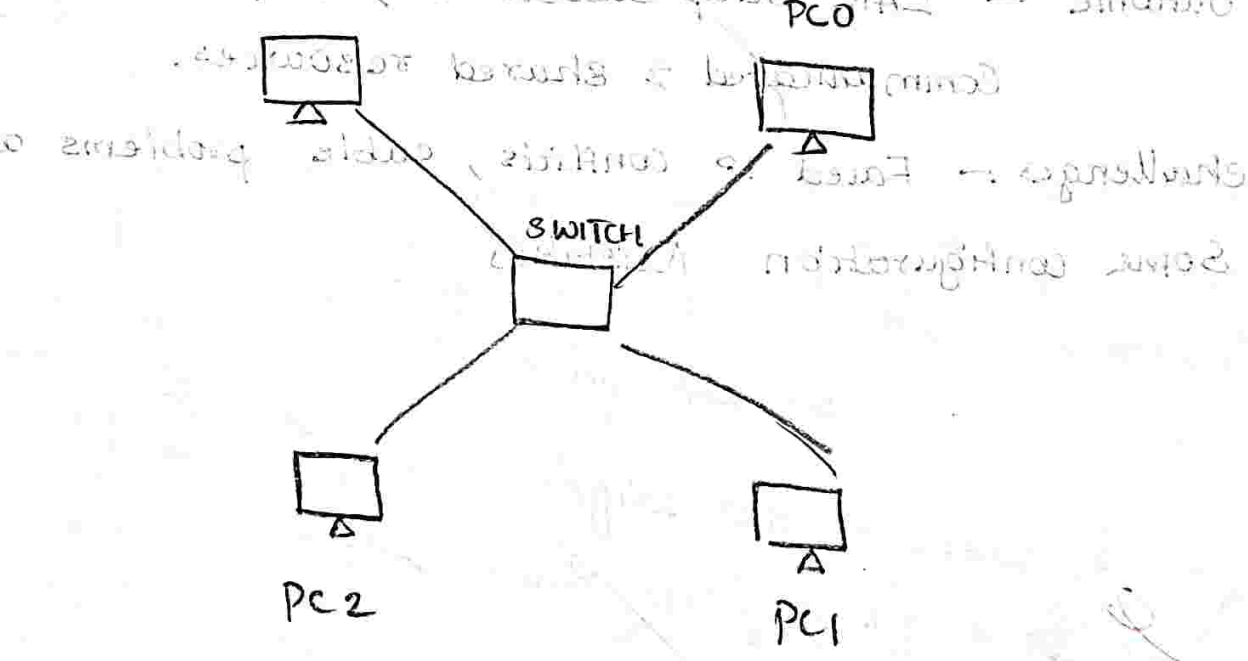
- Assign Switch IP : 10.1.1.5, Subnet Mask : 255.0.0.0

Step 5 : Test Connectivity

- Open Command Prompt on any PC
- Type : ping 10.1.1.5 (to check switch) or ping 10.1.1.2 (to check other PC)

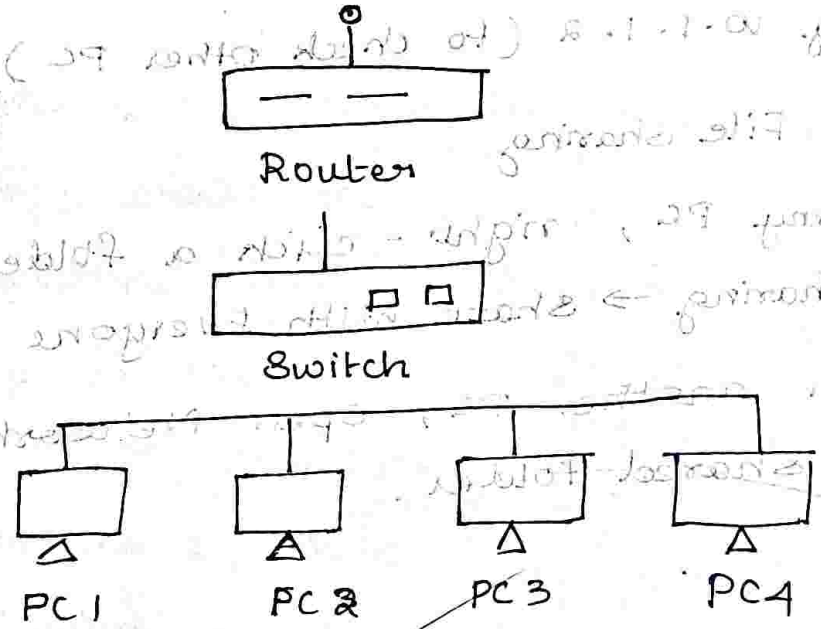
Step 6 :- File Sharing

- On any PC, right-click a folder → properties → Sharing → Share with Everyone.
- From another PC, open Network to access the ~~Shared~~ folder.



Student Observation

① Draw neat diagram of the LAN implemented in the Write IP Config of Each device, & mention the outcome and challenges faced.



Outcome — LAN Setup Successful ; all devices Communicated & shared resources.

challenges :- Faced IP conflicts, cable problems and some configuration Mistakes

Result :- Setup & configure a LAN using a Switch and Ethernet Cable is done Successfully.

15/9/25

Exp-8

NMAP on the TryHackme platform.

AIM :-

This Experiment Outline the processes that Nmap takes before port-scanning to find which system are outline

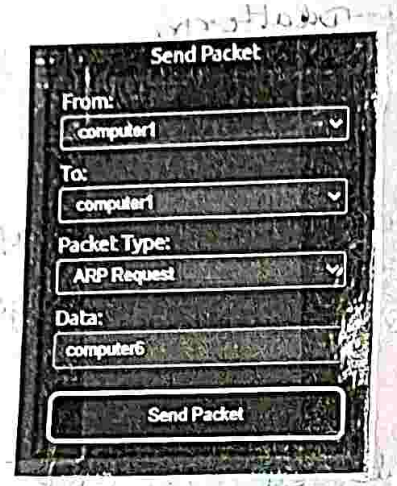
1) ARP Scan : Uses ARP requests to discover live hosts
2) ICMP Scan : This scan uses ICMP requests to identify live hosts
3) TCP / UDP Ping scan : Sends packets to TCP ports and UDP ports to determine live hosts

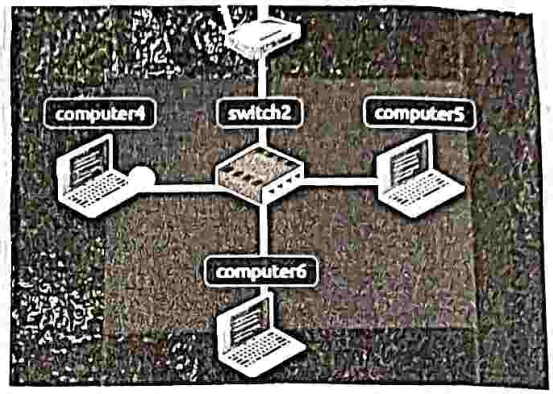
There will be 2 Scanners introduced

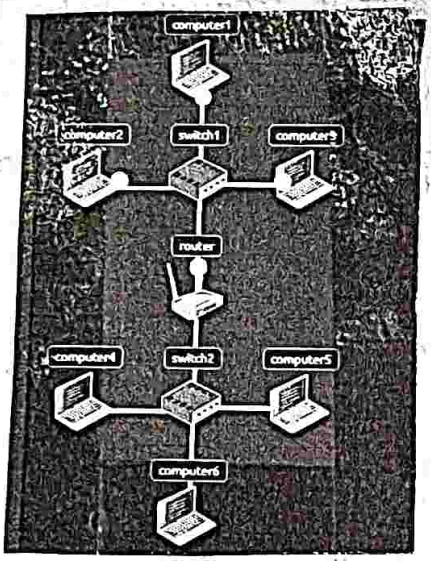
1) arp-scan
2) mass can

Nmap (Network Mapper) - It is a well-known tool for mapping Networks, locating live hosts and detecting running services.

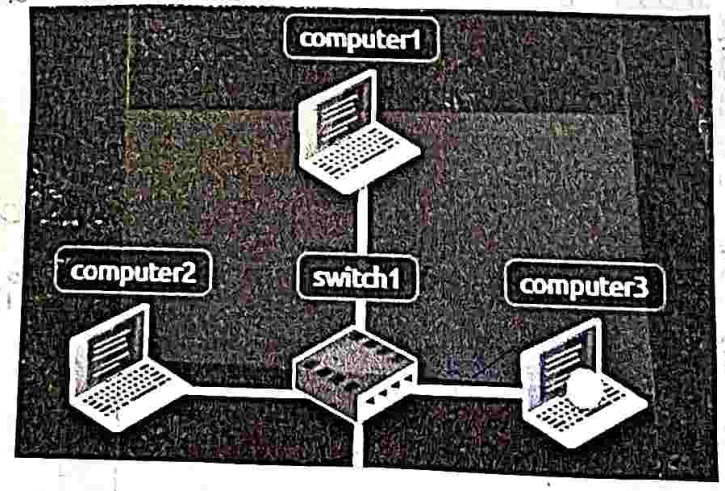
Nmap's scripting engine can be used to Extend its capabilities such as finger printing services and Exploiting flaws.

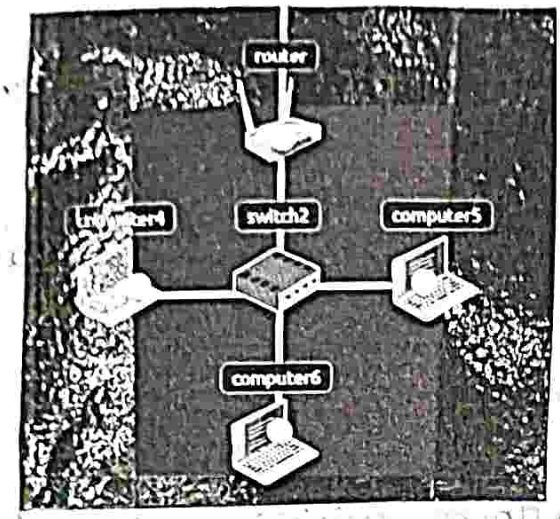

Send Packet
From:
computer1
To:
computer1
Packet Type:
ARP Request
Data:
computer6
Send Packet

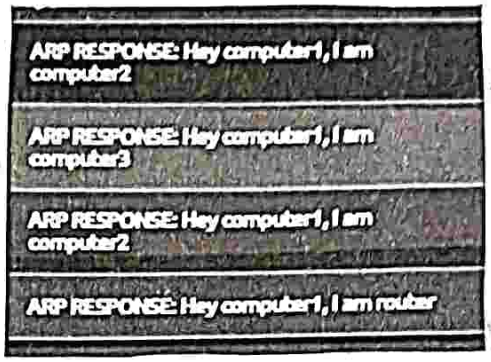

computer4
switch2
computer5
computer6

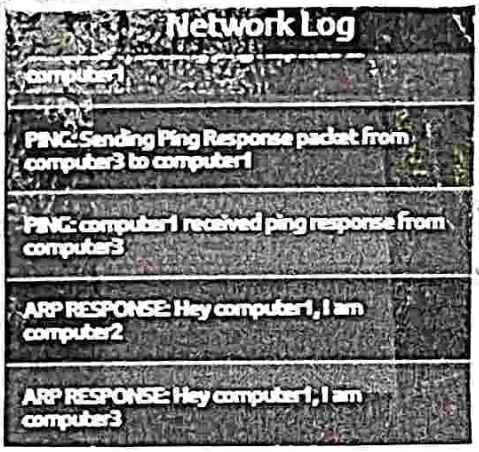

computer1
computer2
switch1
computer3
router
computer4
switch2
computer5
computer6

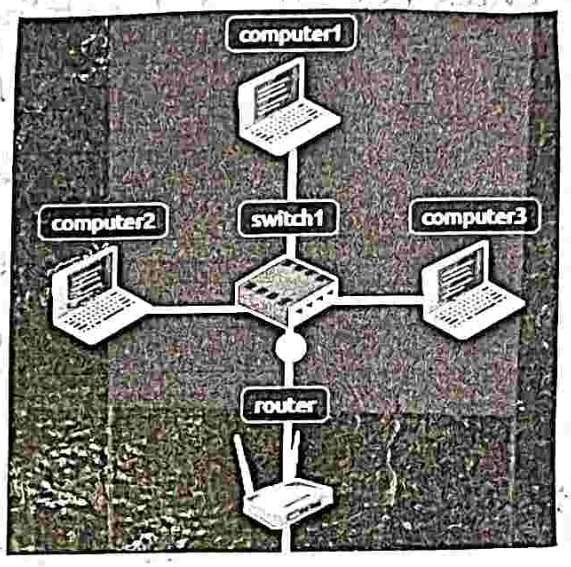

Send Packet
From:
computer4
To:
computer4
Packet Type:
ARP Request
Data:
computer6
Send Packet

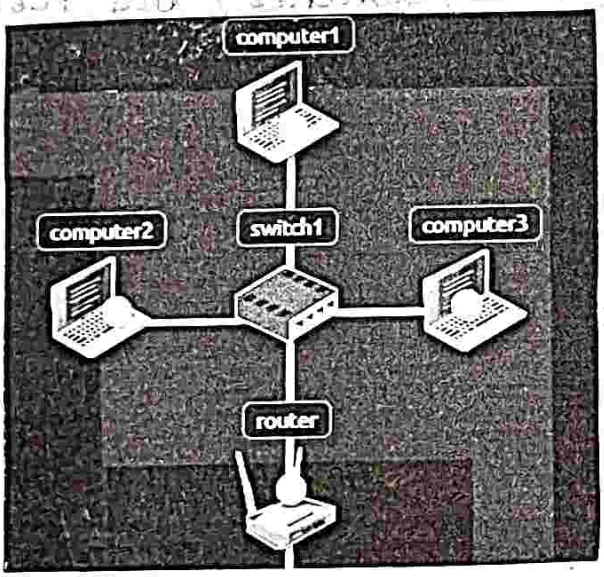

computer1
computer2
switch1
computer3

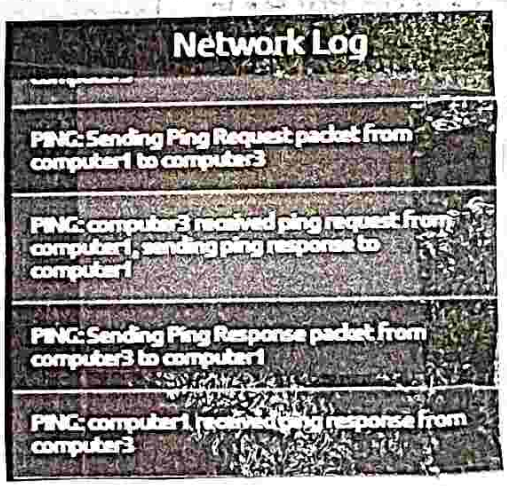












RESULT :-

The Experiment NMAP on the Tryhackme platform is done successfully.

8/10/21

29/9/25

Exp-9

SUBNETTING In CISCO PACKET TRACER.

AIM :

Implementation of Subnetting in Cisco packet Tracer Simulator

CONCEPT :

classless IP subnetting allows dividing a network into smaller subnets for efficient IP ~~address utilizat~~ usage. Instead of fixed classful Masks, subnetting uses variable-length subnet masks (VLSM) to create networks as per requirements

STEPS :-

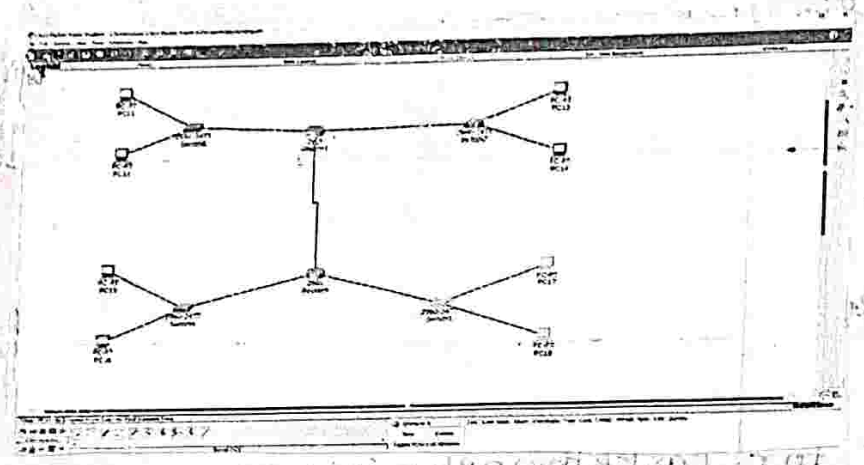
1. Create Topology : Add routers, switches, and PCs in Cisco Packet Tracer.
2. Subnetting : Example - 192.168.1.0/24 subnetted into /27 gives 8 subnets ~~IP to routers, switches, and PCs.~~ with 30 hosts each.
3. Assign IPs : Allocate subnet IPs to routers, switches, and PCs. Switch 1 & 2 = FA0/1 : 192.168.1.0/27
   PCs :- 192.168.1.11-15
   Routers = Gig0/0 : 192.168.1.1
4. Configure Devices :

   Router → Configure interfaces with IP & subnet mask.

   Switch → set access mode on ports.

PCs → Assign IP, Subnet Mask, and default gateway.

5. Test : Use ping to check connectivity among devices.



STUDENT OBSERVATION :

a) write down your understanding of Subnetting ?

Ans :- Subnetting divides a big network into small parts for better control.

b) what is the adv of Impl subnetting within a Network?

Ans :- It Improves security, speed, and saves IP address

c) Find out whether subnetting is implemented in your clg.?

Ans Yes colleges use subnetting Example

- Admin : 192-168-1-0/26
- Lab : 192-168-1-64/26

Result :-

Implementation of Subnetting in CISCO PACKET Tracer Simulator is done Successfully.

3/10/25

3/10/25

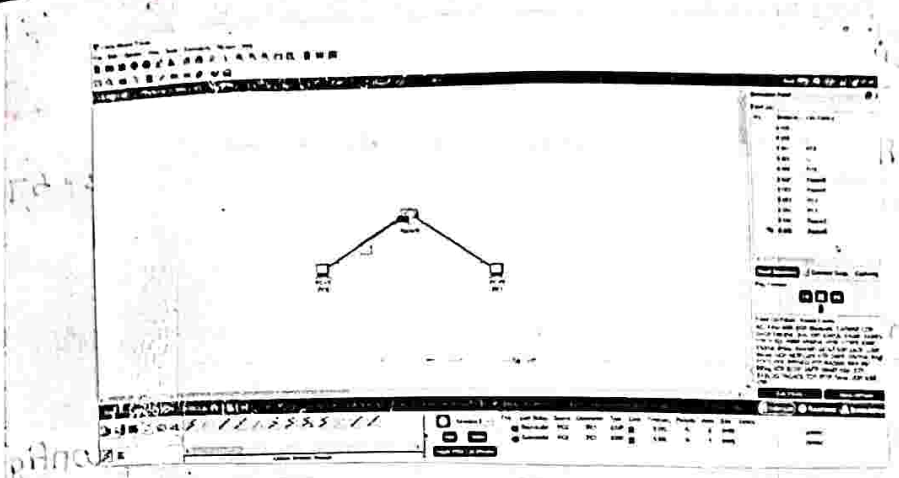
Exp no a) INTERNETWORKING WITH ROUTERS IN CISCO PACKET TRACER SIMULATOR.

a) AIM

Internetworking with routers in CISCO PACKET TRACER Simulator, ~~Design~~ Design a Simple network with 1 router and 2 PCs using Cisco packet tracer

Procedure:

1. Configure Router:
   - > Go to CLI → enable → config t
   - > Set IP for FastEthernet 0/0 : 192.168.10.1
     255.255.255.0 → no shutdown.
   - > Set IP for FastEthernet 0/1 : 192.168.20.1
     255.255.255.0 → no shutdown.
2. Configure PCs:
   - > PC0 : IP 192.168.10.2, Gateway 192.168.10.1
   - > PC1 : IP 192.168.20.2, Gateway 192.168.20.1
3. Connect Devices:
   - > PC0 → Router FastEthernet 0/0 (copper straight-through)
   - > PC1 → Router FastEthernet 0/1 (copper straight-through)
4. Test Connectivity:
   - > ping from PC0 to PC1 to verify network connection



b) AIM:

Configure a Network with a wireless router, DHCP server and Internet connection.

Procedure:

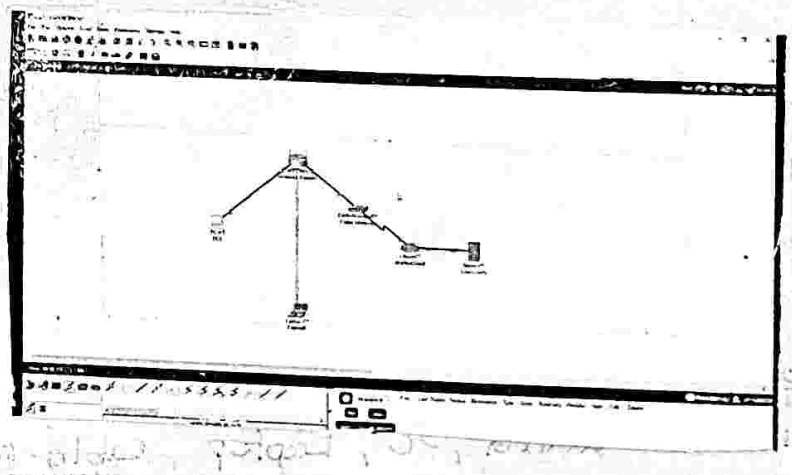
1. Build Network:
   - > Add Wireless Router, PC, Laptop, Cable Modem, Internet Cloud, Cisco.com server.
   - > Connect using copper and coaxial cables as required.
2. Configure Wireless Router.
   - > Set SSID: HomeNetwork.
   - > Enable DHCP & set DNS: 208.67.220.220.
3. Configure Laptop & PC:
   - > Laptop: Install wireless module → Connect to Home Network.
   - > PC: Enable DHCP → Get IP automatically from router.

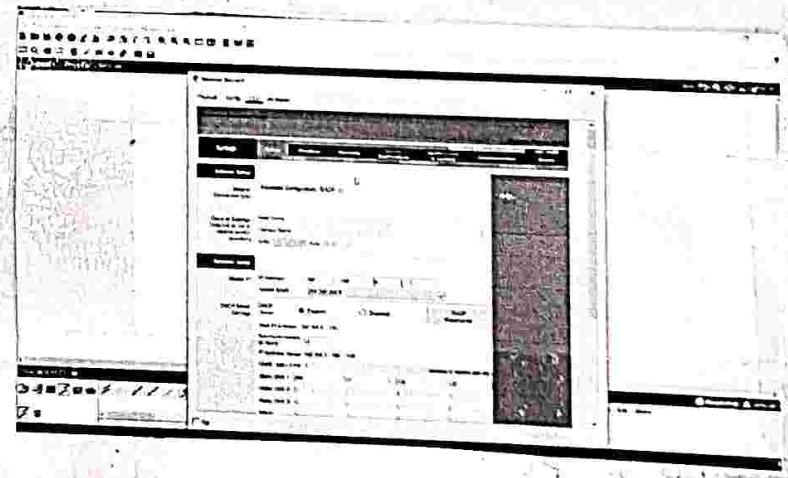
4. Configure Cisco.com Server :

> Enable DHCP & DNS services with IP : 208.67.220.22
> Set pool & global gateway settings.

5. Verify Connectivity :

> Refresh IP on PC (ipconfig /release → ipconfig /renew
> Ping cisco.com to test internet connection





STUDENT OBSERVATION

1. Write down the key features of configuring wireless router & DHCP server.

Ans Wireless Router : Provides wi-fi, sets SSID, connects devi

DHCP Server : Automatically assigns IP, gateway, subnet, DNS.

2. What is the significance of DHCP Server in internetwork-ing ?

Ans:
- Simplifies IP assignment & avoids conflicts
- Ensures smooth network communication.

3. Design & Configure an inter-network in your lab using switch, router & Ethernet cables. Draw & label.

Ans:
> Setup : Router → Switch → PCs (Ethernet cables)
> Eg:-
- Router : 192.168.1.1
- PC0 : 192.168.1.2 , Gateway : 192.168.1.1
- PC1 : 192.168.1.3 , Gateway : 192.168.1.1

> Test : ping between PCs to verify connection.

Result:
Internetworking with routers in cisco packet Tracer simulator is done successfully.

9/10/25

5/10/25

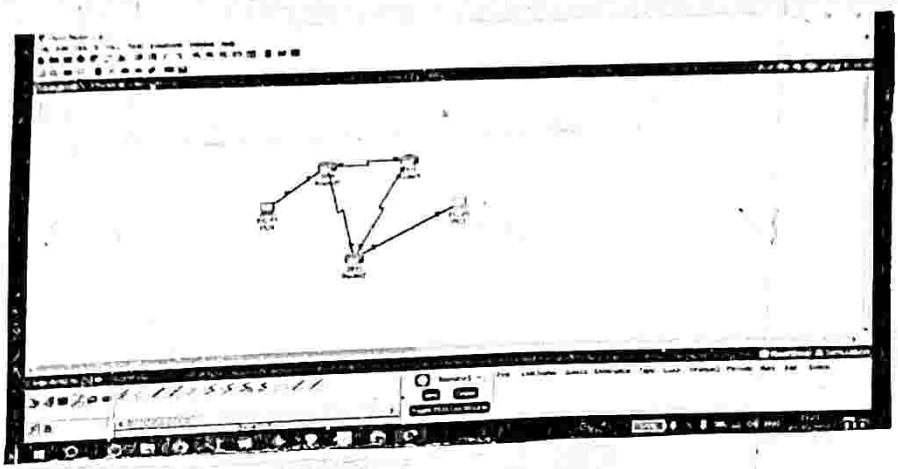
Exp: 11 SIMULATE STATIC ROUTING CONFIGURE USING CISCO Packet Tracer.

(a) AIM: Static Routing Configuration

To simulate static routing configuration using Cisco packet Tracer & understand how to Manually add, Verify, and Manage static routes in a Network

PROCEDURE

1. Open Cisco packet Tracer & create 3 routers (R0, R1, R2) with PCs.
2. Connect router and PCs using proper cables
3. Assign IP address to all Interfaces based on the given network table.
4. Configure static routes on each router for Network not directly Connected.
5. Verify routing table to check main and backup ro
6. Test connectivity using ping & tracert b/w PCs
7. Simulate link failure by removing a connection and check backup route.
8. Delete static route if needed using router CLI

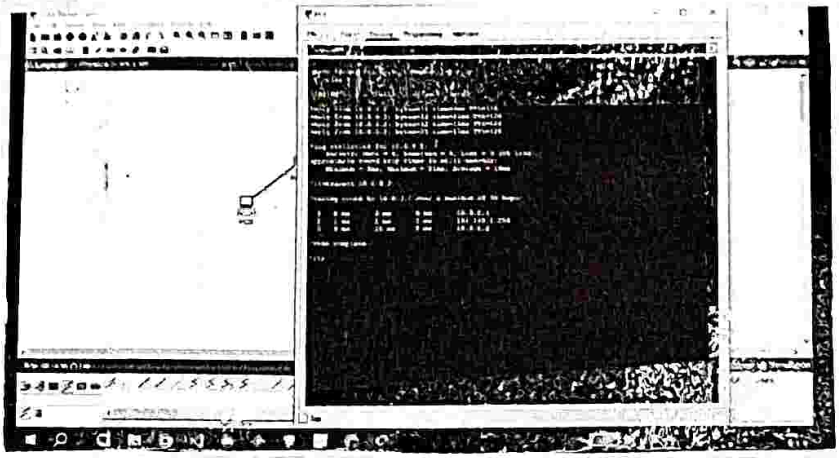


b) AIM : RIP Configuration

To Simulate Routing Information protocol (RIP) using Cisco packet Tracer & understand how routers automatically share routing information within a network.

PROCEDURE

1. Create a Network using 3 routers (R0, R1, R2) & 2 PCs.
2. Connect all devices & assign IP addresses as per table.
3. Enable all Interfaces of routers.
4. Enable RIP protocol on all routers & add network addresses.
5. check routing tables entries learned through RIP
6. Verify network connectivity using ping btw PCs.



Result :

Simulate Static Routing Configuration using Cisco packet Tracer is done Successfully.

8/10/24

7/10/25

Exp:-12.

Practical 12

AIM:-

a) Implement Echo client Server using TCP/UDP Sockets

Algorithm :-

```
import time
def ping_server(host='127.0.0.1', port=12345):
    with Socket.& Socket(Socket.AF_INET, Socket.SOCK_
        DGRAM) as s:
        try:
            s.sendto(b'Hello', (host, port))
        Except s.timeout:
            print("Request timed Out")
ping_server()
```

Result :-

The Message hello is Successfully sent to the Server.

9/10/25

b) AIM:- Implement chat client server using TCP / UDP sockets.

Algorithm :-

```
import socket
def start_server(host = '127.0.0.1', port=12345):
    with socket.socket(socket.AF_INET, socket_sock
                       DGRAM) as s:
        s.bind((host, port))
        print(f"UDP Server running on {host}:{por

        data, addr = s.recvfrom(1024)
        print(f"Received Message from {addr}
                                {data.decode()}"

start_server()
```

OUTPUT :

```
UDP Server running on 127.0.0.1 : 12345
Received Message from ('127.0.0.1', 52345): Hel
```

8/10/25

Exp: 13

Practical 13

AIM:

Implement your own ping program.

ALGORITHM:

1. Create a UDP socket.
2. Set a timeout of 2 seconds
3. Record start time
4. Send the message "ping" to the server.
5. Wait to receive a response from the server
   - > If received, record end time and display the reply with round-trip time.
   - > If timeout occurs, print "Request time out".
6. close the socket.

INPUT:

```
import socket
import time
def ping_server(host='127.0.0.1', port=12345):
    with
socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
try:
    s.settimeout(2)
    start = time.time()
```

```
        s.sendto(b'Ping', (host, port))
        data, addr = s.recvfrom(1024)
        end = time.time()
        print(f"Received {data.decode()} from
            {addr} in {end-start:.2f} seconds")
    except socket.timeout:
        print("Request timed out")

if __name__ == "__main__":
    ping_server()
```

OUTPUT:

```
Received Pong from ('127.0.0.1', 12345) in 0.00 seconds
Request timed out
```

Result:

Successfully Implement chat client Server using TCP / UDP sockets

9/10/24

b) Algorithm :

1. Create a UDP socket
2. Bind the socket to server IP and port
3. Continously wait for incoming Messages.
4. When a message is received :
   > Display the Message and client address.
   > Send back "pong" to the client.
5. Repeat step 3 indefinitely.

INPUT:

```
import socket

def start_server(host='127.0.0.1', port=12345):

    with
socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
as s:
        s.bind((host, port))
        print(f"UDP Server running on {host}:{port}")

    while True:
        data, addr = s.recvfrom(1024)
        print(f"Received message from {addr}:
                {data.decode()}")
        s.sendto(b"Pong", addr)
```

```
if __name__ == "__main__":
    start_server()
```

O/p:-

```
UDP Server running on 127.0.0.1:12345
Received message from ('127.0.0.1', 52348): ping
```

Result:

Ping program is implemented Successfully

9/10/25

Exp: 14

Practical 14

AIM:

Write a code using RAW sockets to implement packet sniffing

ALGORITHM:

1. Open a raw socket in promiscuous mode
2. Receive a packet (raw bytes) from the socket.
3. Parse Ethernet header to get EtherType
4. If EtherType == IPV4, parse IP header to get protocol, src IP, dst IP.
5. Optionally parse TCP / UDP / ICMP headers based on IP protocol
6. Display / log the packet info. Repeat to keep sniffing.

INPUT:

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
```

```
# determine the protocol.
protocol_name = " "
if protocol == 1:
    protocol_name = " ICMP"
elif protocol == 6:
    protocol_name = " TCP"
elif protocol == 17:
    protocol_name = " UDP"
else:
    protocol_name = " Unknown protocol"
# print packet details.
print(f" protocol : {protocol_name}")
print(f" Source IP: {src_ip}")
print(f" Destination IP: {dst_ip}")
print("_" * 50)
# capture packets on the default network interface.
sniff(iface = 'Wi-Fi', prn=packet_callback,
      filter = "ip", store=0)
```

OUTPUT:

protocol : TCP

Source IP: 192.168.1.5

Destination IP: 172.217.16.142

Protocol : ICMP

Source IP: 192.168.1.2.

Destination IP: ~~224.0.0.251~~ 8.8.8.8.

---

protocol : UDP

Source IP : 192.168.1.10.

Destination IP : ~~172~~ 224.0.0.251

---

protocol : TCP

Source IP : 192.168.1.5

Destination IP : 172.217.16.142

Result:

Successfully executed code using RAW Sockets to implement packet sniffing.